Date:28.02.2026	**Logistic Regression**
Ex. No: 5	

AIM

To build a Logistic Regression model to detect the presence of breast cancer using Logistic Regression and evaluate its performance using classification metrics and ROC-AUC analysis.

The objective is:

- To classify breast cancer as Benign (0), Malignant (2)
- To minimize false negatives (missing Malignant cases)
- To analyze model performance using appropriate evaluation metrics

ALGORITHM USED: Logistic Regression

Logistic Regression is a supervised binary classification algorithm used when the target variable has two classes.

Algorithm Logistic_Regression (X_train, X_test, y_train, y_test) return success or failure

X_train : (x1,x2,…,xn training features)

X_test: (diagnosis test feature)

y_train: (x1,x2,…,xn testing features)

y_test: (diagnosis test feature)

Import necessary libraries

Checking for null values using df.isnull().sum()

Encode the diagnosis column using Label Encoder()

Computing a linear combination:

$$z = w_1x_1 + w_2x_2 + \ldots + w_nx_n + b$$

Applying the Sigmoid Function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Converting probability into class:

- If probability ≥ 0.5 → Class 1 (Malignant)
- If probability < 0.5 → Class 0 (Benign)

Probability:

- P(y=1) = $\sigma(z)$
- P(y=0) = 1- $\sigma(z)$

return accuracy_score

Train-Test Split

- 80% Training data
- 20% Testing data
- random_state=42 for reproducibility

Feature Scaling

- Used StandardScaler
- Important because Logistic Regression is sensitive to feature scale

EVALUATION METRICS

1.Accuracy

$$\boldsymbol{Accuracy = (TP + TN)/(TP + TN + FP + FN)}$$

2. Precision

$$\boldsymbol{Precision = TP/(TP + FP)}$$

3.Recall (Sensitivity)

$$\boldsymbol{Recall = TP/(TP + FN)}$$

4.F1-Score

$$\boldsymbol{F1 = 2 * (Precision * Recall)/(Precision + Recall)}$$

5. ROC Curve

ROC plots:

- **X-axis → False Positive Rate**
- **Y-axis → True Positive Rate**

6. AUC (Area Under Curve)

$$\boldsymbol{AUC = AreaunderROCcurve}$$

CODE:

```
import pandas as pd
df=pd.read_csv(r'/content/breast_cancer_data.csv')
df.head()
```

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_mean
0	842302	M	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710
1	842517	M	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017
2	84300903	M	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790
3	84348301	M	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520
4	84358402	M	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430

5 rows × 32 columns

```
df.shape
```

(569, 32)

df.isnull().sum()

	0
id	0
diagnosis	0
Radius_mean	0
Texture_mean	0
perimeter_mean	0
area_mean	0
smoothness_mean	0
compactness_mean	0
concavity_mean	0
concave points_mean	0
symmetry_mean	0

THERE ARE NO NULL VALUES

df['diagnosis'].value_counts()

diagnosis	count
B	357
M	212

dtype: int64

```
from sklearn.preprocessing import LabelEncoder
le=LabelEncoder()
df['diagnosis']=le.fit_transform(df['diagnosis'])
X=df.drop('diagnosis',axis=1)
y=df['diagnosis']

from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
X=scaler.fit_transform(X)
```

```
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2)

from sklearn.linear_model import LogisticRegression
lr=LogisticRegression()
lr.fit(X_train,y_train)
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
y_pred=lr.predict(X_test)

acc=accuracy_score(y_test,y_pred)
class_report=classification_report(y_test,y_pred)
cm=confusion_matrix(y_test,y_pred)

print("The Accuracy is: ",acc)
print("Classification Report: ")
print(class_report)
print("Confusion Report")
print(cm)
import seaborn as sns
import matplotlib.pyplot as plt
sns.heatmap(cm,annot=True)
plt.title("Confusion Matrix")
```

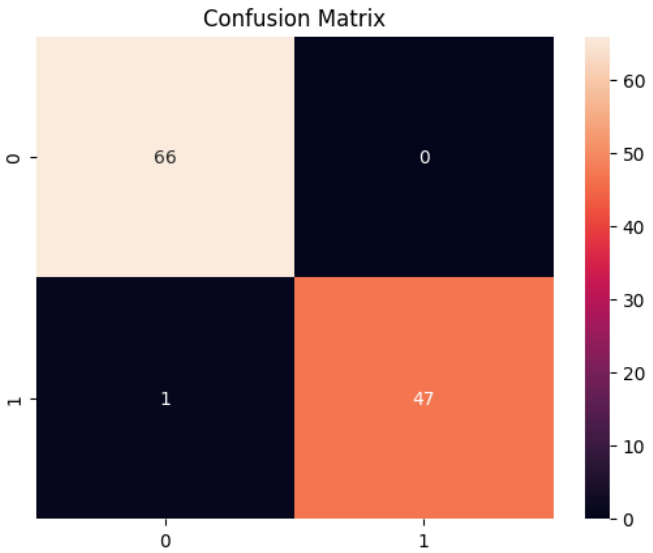
RESULTS:

```
The Accuracy is:  0.9912280701754386
Classification Report:
              precision    recall  f1-score   support

           0       0.99      1.00      0.99        66
           1       1.00      0.98      0.99        48

    accuracy                           0.99       114
   macro avg       0.99      0.99      0.99       114
weighted avg       0.99      0.99      0.99       114

Confusion Report
[[66  0]
 [ 1 47]]
```



Logistic Regression at Multi-class Classification:

Results:

fetal_health	count
1.0	1655
2.0	295
3.0	176

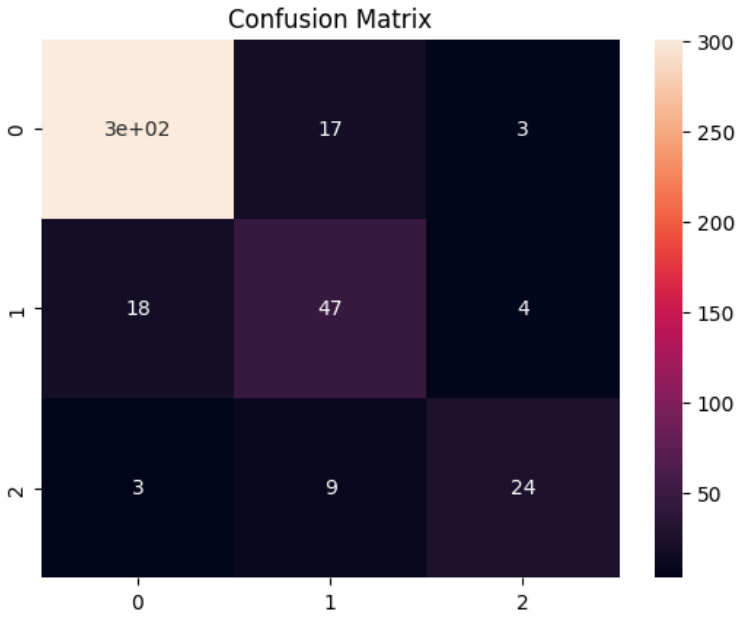
dtype: int64

```
The Accuracy is:  0.8732394366197183
Classification Report:
              precision    recall  f1-score   support

         1.0       0.93      0.94      0.94       321
         2.0       0.64      0.68      0.66        69
         3.0       0.77      0.67      0.72        36

    accuracy                           0.87       426
   macro avg       0.78      0.76      0.77       426
weighted avg       0.87      0.87      0.87       426

Confusion Report
[[301  17   3]
 [ 18  47   4]
 [  3   9  24]]
```



INFERENCE:

Thus we can observe from the above results that Logistic Regression is working efficiently for binary class dataset and struggling for multi-class dataset due to its probabilistic nature.